MQTT and IoT Platforms

Advanced Embedded Software Development

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Learning objectives:

Introduce the MQTT protocol

Understand how to use security with MQTT based projects.

Introduce the JSON format.

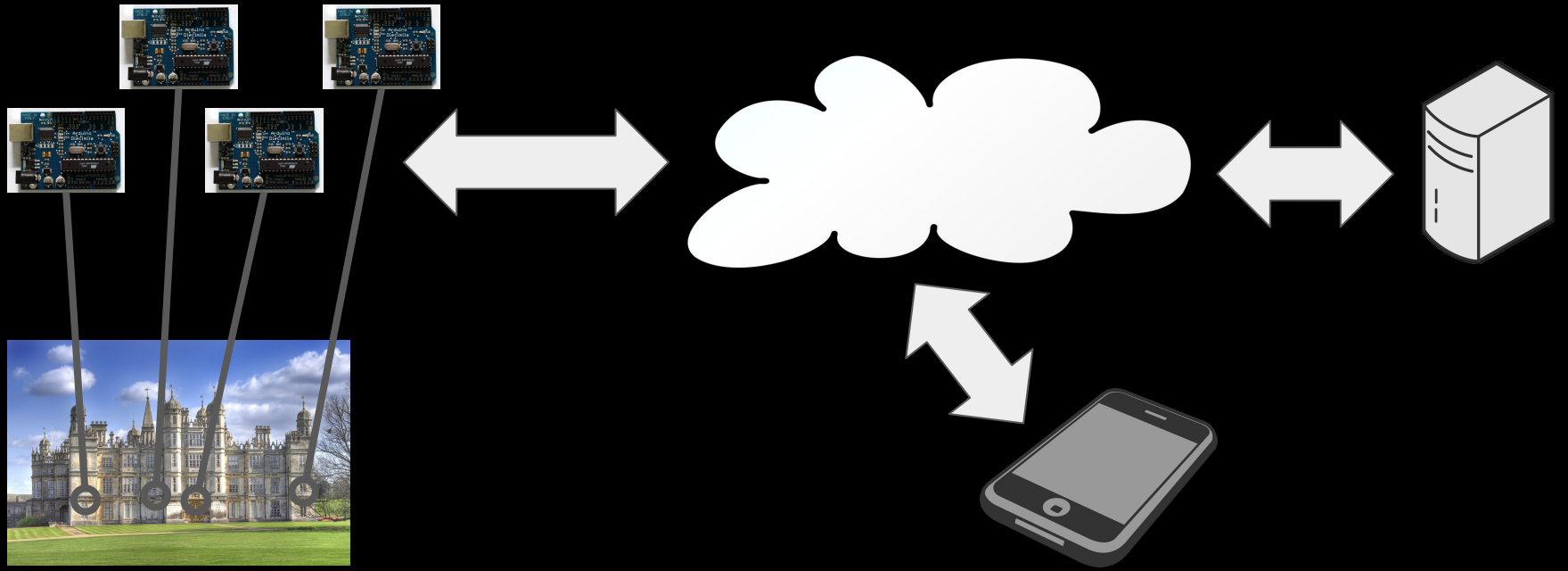
Introduce Device Shadows/Digital Twins

MQTT

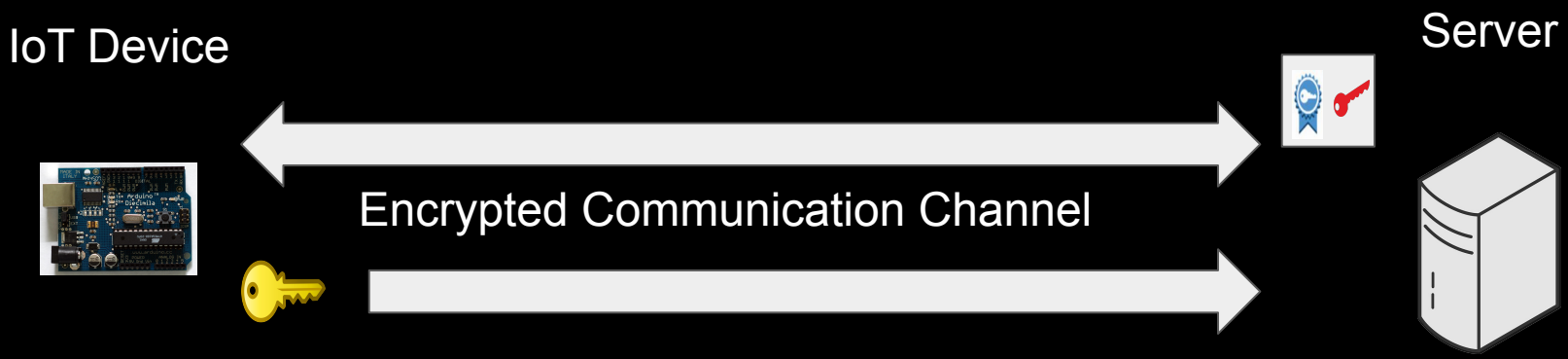
- Message Queueing Telemetry Transport (1999)
- Lightweight publish/subscribe messaging transport
  - Low bandwidth
  - Efficient distribution to one or many receivers
- Popular with IoT platforms
- Can be securely transmitted using SSL/TLS
    - Protocols for establishing and authenticating encrypted links between devices
      - Same technology behind https://
- Mosquitto client/server open source software available at https://mosquitto.org/

http://mqtt.org/
https://mqtt.org/faq

MQTT IoT Example



Basic Security Scheme



- Secure Bidirectional Pipe with Server Identity Verified by Certificate Authority signed certificate.
- Client Identified Through Encrypted Secret Transmission of Private Key or Password.
  - Key transfer relies on security of encrypted channel.
  - A hacker able to obtain the key (man in the middle) can emulate the Device with the same permissions on the Server

MQTT SSL Security Scheme

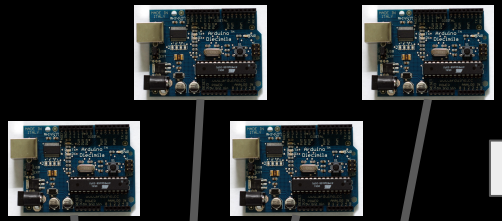


- Secure Bidirectional Pipe with Server And Device Identity Verified by Certificate Authority signed certificate.
- No Private Key or Password Transmission
  - A hacker who can compromise the channel can view/modify data sent between device and server but cannot emulate the device.

JSON Format

- JavaScript Object Notation
- Developed in ~1999 as a part of the JavaScript Programming Language Standard
- Language independent
- Text based
- object - ordered set of name/value pairs, surrounded by braces
- Object name followed by colon, name/value pairs separated by comma
- Easy to parse by embedded software

Amazon IoT example MQTT Shadows

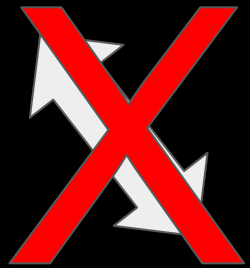


Door = open



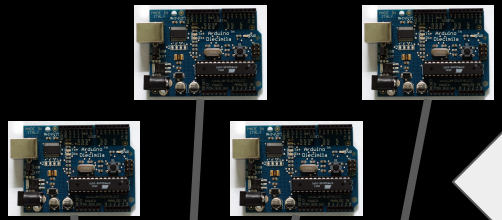


{ “state”: { “reported” : { “frontdoor” : “open” } } }





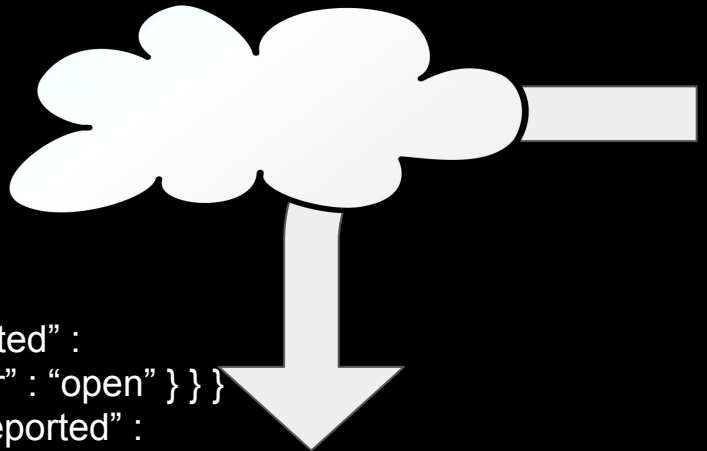
Amazon IoT example MQTT Shadows



Door = open







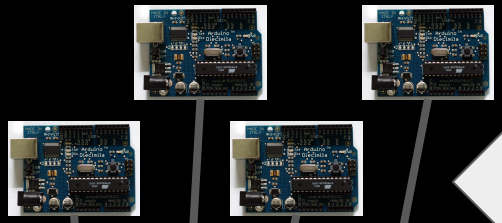
AWS IoT



```
{ “state”: { “reported” :
        { “frontdoor” : “open” } } }
{ “metadata: { “reported” :
        { “frontdoor” : { “timestamp” : 1452964452 } } } }
```



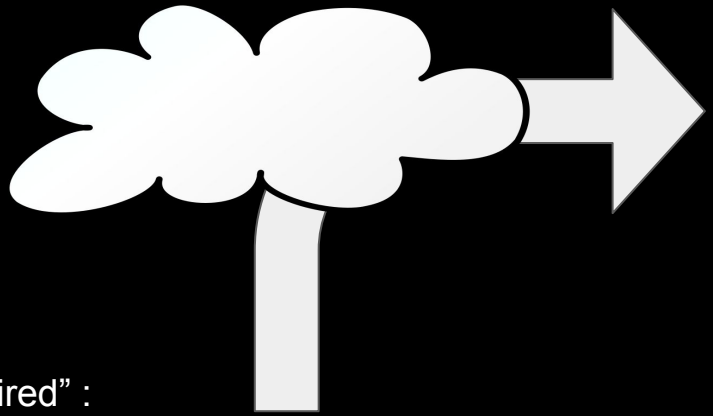
Amazon IoT example MQTT Shadows



Door = open







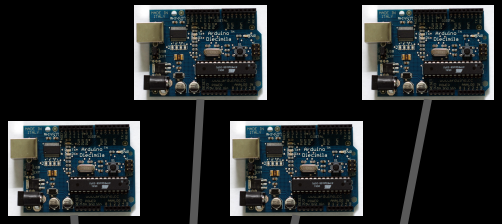
AWS IoT



{ “state”: { “desired” :
{ “frontdoor” : “close” } } }



Amazon IoT example MQTT Shadows



Door = open



AWS IoT





{ “state”: { “desired” :
{ “frontdoor” : “close” } } }
{ “state”: { “delta” :
{ “frontdoor” : “close” } } }



Amazon IoT example MQTT Shadows



AWS IoT

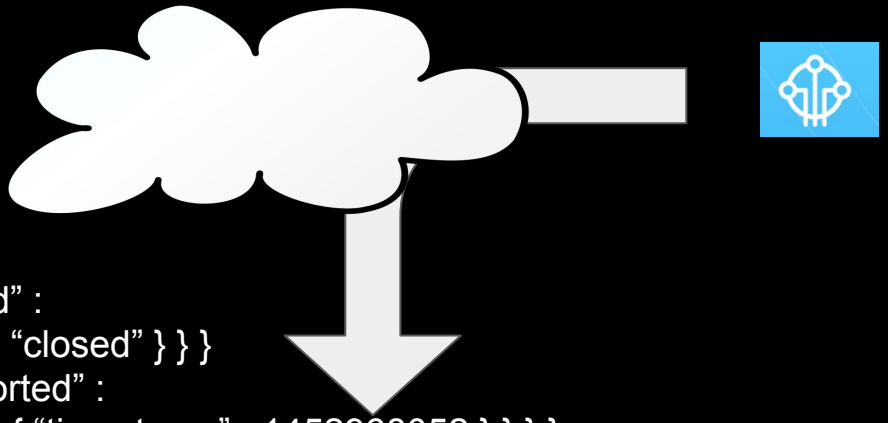
{ “state”: { “reported” :
{ “frontdoor” : “closed” } } }



Amazon IoT example MQTT Shadows



AWS IoT



```
{ “state”: { “reported” :
        { “frontdoor” : “closed” } } }
{ “metadata: { “reported” :
        { “frontdoor” : { “timestamp” : 1452968052 } } } }
```



Amazon IoT for Embedded Links

- Getting Started with AWS IoT
- AWS IoT SDK Tutorials
  - Includes example for Raspberry Pi and Raspbian
  - You could also start by running on your Ubuntu VM
- AWS IoT Embedded C SDK
- Similar services:
  - Google Cloud IoT (Device Registry)
  - IBM Watson IoT (Digital Twins)
  - Microsoft Azure IoT (Device Shadows)
    - See student implementation at https://github.com/cu-ecen-5013/yocto-stm32mp1/wiki/Azure-IoT-Setup

https://developer.ibm.com/technologies/iot/articles/digital-twins-and-the-internet-of-things/